

Practice Problems

Problem 2.1

An investment of \$1,000 grows by a constant amount of \$250 each year for five years.

1. What does the graph of $A(t)$ look like if interest is only paid at the end of each year?
2. What does the graph of $A(t)$ look like if interest is paid continuously and the amount function grows linearly?

Solution

Since the investment increases by a constant amount of \$250 each year, the account values at the end of each year are:

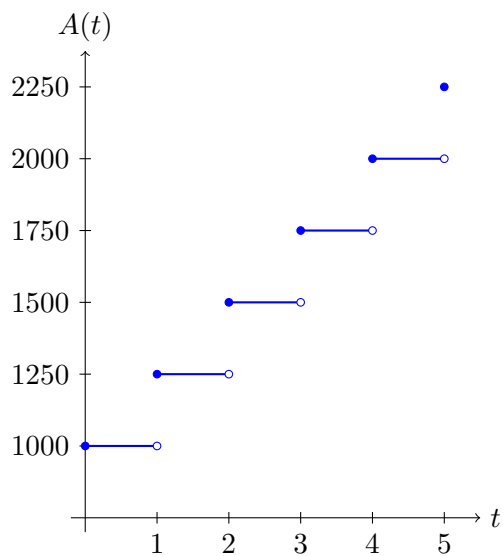
t	1	2	3	4	5
$A(t)$	1250	1500	1750	2000	2250

(a) If interest is only paid at the end of each year, then the graph of $A(t)$ is a *step function*. The amount stays constant during each year and jumps up by \$250 at the end of each year.

A convenient way to describe it is:

$$A(t) = \begin{cases} 1000, & 0 \leq t < 1, \\ 1250, & 1 \leq t < 2, \\ 1500, & 2 \leq t < 3, \\ 1750, & 3 \leq t < 4, \\ 2000, & 4 \leq t < 5, \end{cases} \quad A(5) = 2250.$$

The corresponding graph is:



(b) If interest is paid continuously and the amount function grows linearly, then the graph is a straight line joining the points from the initial amount to the final amount.

The linear amount function is

$$A(t) = 1000 + 250t, \quad 0 \leq t \leq 5.$$

This line starts at $A(0) = 1000$ and ends at $A(5) = 2250$.

The corresponding graph is:

